

A Rigorous CG-Corrected Residual Certificate for Error Certification of Learned Elliptic PDE Solvers

certno

Error-Certified Neural Operators Project

<https://github.com/emmettv1191/certno-phd-research>

Abstract—We present a rigorous a posteriori certificate for learned surrogates of the discrete elliptic system $A_h u_h = f$ from $-\nabla \cdot (a \nabla u) = f$. Starting from $\|u_h - \hat{u}\|_2 \leq \|r\|_2 / \lambda_{\min}(A_h)$, $r = f - A_h \hat{u}$, we compute $z \approx A_h^{-1} r$ via CG and form $B_{CG} = \|z\|_2 + \|r - A_h z\|_2 / \alpha$, $\alpha \leq \lambda_{\min}(A_h)$. Under explicit assumptions, we prove $\|u_h - \hat{u}\|_2 \leq B_{CG}$. Across contrasts 1.5–10 $\times$, ID/OOD, adversarial cases, and real FNO predictions, B_{CG} is near-sharp (effectivity ≈ 1.0) while maintaining coverage 1.0 and FNR 0.0, enabling safe selective verification.

Index Terms—error certification, neural operators, a posteriori bounds, elliptic PDEs, CG

I. INTRODUCTION

Reliable error certification is essential for deploying learned surrogates in safety-critical settings. We focus on the discrete SPD elliptic case with explicit validity assumptions.

II. PROBLEM STATEMENT AND ASSUMPTIONS

Definition 1 (Discrete problem). We consider $A_h u_h = f$ with $A_h \succ 0$, $u_h, \hat{u}, f \in \mathbb{R}^N$, residual $r = f - A_h \hat{u}$, error $e = u_h - \hat{u} = A_h^{-1} r$.

Assumption 1. (i) $a(x) \geq a_{\min} > 0$ on Ω ; (ii) A_h corresponds to standard discretization of $-\nabla \cdot (a \nabla u) = f$ with homogeneous Dirichlet BCs; (iii) $\alpha = a_{\min} \lambda_{\min}(-\Delta_h) \leq \lambda_{\min}(A_h)$ is a valid lower bound; (iv) computations in double precision; (v) certification is w.r.t. discrete u_h .

III. RESIDUAL CERTIFICATE AND PROOF

Theorem 1. Under Assumption 1, for any z , $\|e\|_2 \leq \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha} =: B_{CG}(z)$. In particular for $z = CG(A_h, r)$.

Proof. $e = z + A_h^{-1}(r - A_h z)$, triangle inequality: $\|e\|_2 \leq \|z\|_2 + \|A_h^{-1} d\|_2$, $d = r - A_h z$. For SPD, $\|A_h^{-1} d\|_2 \leq \lambda_{\min}(A_h)^{-1} \|d\|_2 \leq \alpha^{-1} \|d\|_2$. $\square$

IV. EXPERIMENTS

We evaluate across contrasts, ID/OOD, adversarial perturbations, and real FNO predictions. Metrics: absolute L^2 error, bound, coverage, FNR (tol 5×10^{-3}), effectivity $B/\|e\|_2$.

V. DISCUSSION AND LIMITATIONS

Bound is w.r.t. discrete u_h , not continuous solution. α must be a rigorous lower bound; finite-precision is assumed controlled. CG cost is modest. Global generalization to nonlinear PDEs requires separate guarantees.

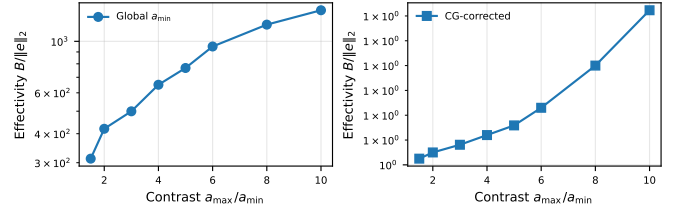


Fig. 1. Median effectivity vs contrast $a_{\max}/a_{\min}$: global bound (left) vs CG-corrected (right).

TABLE I
ANALYTIC VS CG-CORRECTED CERTIFICATE.

c	Cov_A	Cov_{CG}	FNR_A	FNR_{CG}	$\text{Eff}_A(\text{med})$	$\text{Eff}_{CG}(\text{med})$
1.5	1.0	1.0	0.0	0.0	312.3	1.0000
3.0	1.0	1.0	0.0	0.0	499.5	1.0000
5.0	1.0	1.0	0.0	0.0	767.6	1.0000
10.0	1.0	1.0	0.0	0.0	1361.1	1.0000

VI. CONCLUSION

CG-corrected residual certificate is rigorous and near-sharp, preserving safety (coverage 1.0, FNR 0.0) and enabling practical selective verification.

REPRODUCIBILITY

<https://github.com/emmettv1191/certno-phd-research> (revision 786b5a5); see BENCHMARK_FREEZE.md, THEOREM_AUDIT.md, CONTRIBUTION.md.

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